



Pastoralist intelligence for Isiolo's drylands.

SATELLITE → AI → HERDER'S PHONE · VOICE · USSD · SMS

THE PROBLEM

A herder loses KES 200K in a bad drought cycle.

Satellites see it 2–3 wks before the herder does.

THE SOLUTION

Voice-first intelligence on a 2G phone.

Every 5 days · pasture check · outbound call.

VOICE

call

USSD

*123*8#

SMS

BULA

ARCHITECTURE · 3 services · multi-tenant

HERDER

2G/3G

- Voice
- USSD *123*8#
- SMS keywords

AT

telco

- Voice/SMS APIs
- USSD gateway
- DID pool

ARDALINK

3 svcs

- api · engine · web
- API · Biophysical · UI
- AI writes the brief
- Postgres · multi-tenant

DATA + AI

sat/LLM

- Sat reads the ward
- AI writes the brief
- Answers are saved

THE HERDER JOURNEY

1 DIAL

Herder dials *123*8#

2 PICK

1 = Bula Pesa
2 = Malisho
3 = Ongea

3 REPLY

ArdaLink replies Swahili / English

4 CALL

Outbound voice
AI brief in dialect

5 REPORT

Herder answers
Indicators saved

WHAT'S BUILT · WHAT'S NEXT

3

demo wards

134

tests pass

\$0.12

per herder/mo

6 wks

to first call

4

AT webhooks

NEXT

- AT production key + DID pool
- Earth Engine account

- AI voice stack
- 4 local dialects

